ARC-SPEC-2026-EDU-06: Cross-Domain Substrate Realizations & Mechatronic Hardware Transduction

A Standardization Mapping for Physical Toolpaths, Aerospace Micro-Actuators, and Spaceborne Compute Membranes

Jesse Tuohy (Principal Invariant Architect)
Arcstone Adaptive Science Systems

Status: Archive-Ready / Path A Academic Extension
Governing Invariants:
$C_{\text{ops}} = 0 \quad | \quad \tau_{\text{override}} \leq 11.99\text{ms} \quad | \quad \text{Physical Jitter } (\Delta J) \leq 100\text{ns}$
Target Baseline: Phase 7 Realization (`v1.3.0-exec`)

---

1 Executive Summary & Physical Transduction Rationale

ARC-SPEC-2026-EDU-01 through EDU-05 constructed the theoretical curriculum, automated verification harness, accreditation matrix, machine-readable AI grounding schema, and Coq/Isabelle mechanical safety proofs. However, for Deterministic Admissibility Engineering (DAE) to govern cyber-physical reality, software-level admissibility gates must cleanly map into physical energy transduction.

In mechatronic fabrication, robotics, aerospace micro-actuation, and orbital satellite attitude control, software state transitions directly regulate kinetic energy, electrical current, step clocks, and thermal dissipation. If software execution experiences non-deterministic jitter or state drift, physical systems suffer toolpath gouging, mechanical resonance, or catastrophic thermal runaway.

This specification formalizes ARC-SPEC-2026-EDU-06: a cross-domain physical transduction mapping that translates DAE's 5-Tier Poset Dominance Lattice directly into physical hardware interfaces across three primary domains:

1. Terrestrial Mechatronic Fabrication & Additive Synthesis (`DOM-FAB-01`)
2. Autonomous Mobile Robotics & Heavy Machinery (`DOM-ROB-02`)
3. Orbital & Spaceborne Substrate Operations (`DOM-AERO-03`)

Transduction Mapping Comparison

SOFTWARE LATTICE STATE (Ring 0 / eBPF / Microkernel):
[FAIL (POSIX 40)] $\longrightarrow$ [FREEZE (POSIX 10)] $\longrightarrow$ [PWC (POSIX 12)] $\longrightarrow$ [PASS (POSIX 0)]

PHYSICAL HARDWARE TRANSDUCTION (Mechatronic / Substrate):

[Bare-Metal NMI Pin] $\longrightarrow$ [Step Clock Clamp] $\longrightarrow$ [Current Throttle]
$\Downarrow$
$\longrightarrow$ [Synchronous Step Pulse] (Full PWM Torque)

2 Cross-Domain Physical Transduction Mapping Matrix

To ensure software state bounds correspond strictly to bare-metal actuator limits, Table 1 formalizes the hardware transduction matrix across fabrication, robotics, and aerospace substrates.

Table 1: Cross-Domain Physical Transduction Mapping Table

State	**DOM-FAB-01 (Fabrication)**	**DOM-ROB-02 (Robotics)**	**DOM-AERO-03 (Aerospace)**
FAIL	Open AC Relay (0V), NMI Low	Disengage Inverter, Torque 0Nm	Safepoint Hold, Fire Cold-Gas
FREEZE	Clamp Clocks 0Hz, Hold 150°C	Hold Position, Zero Velocity	Despin Wheels, Lock Gimbals
PWC	Feedrate $-20 \ldots 50\%$	Throttle Max Velocity	Downsample Telemetry Stream
REFUSAL	Drop Corrupt G-Code Lines	Reject Kinematic Waypoints	Drop Unverified Ground Frames
PASS	Execute Step Pulse ($\Delta J \leq 100$ns)	Closed-Loop FOC Torque	Normal Orbital Telemetry

3 Physical Hardware Realization Specs

```
#[no_mangle]
pub fn process_toolpath_step(step: &ToolpathStep, sys_state: &
    SystemState) -> PhysicalAction {
    if sys_state.tick_skew > 50 {
        disable_stepper_clocks();
        set_hotend_temperature_celsius(150);
        return PhysicalAction::FreezeHold(POSIX_REASON_CLOCK_SKEW);
    }

    if sys_state.ac_power_fault_detected {
        fram_write_current_block(sys_state.current_gcode_line);
        open_main_power_relay();
        return PhysicalAction::EmergencyPowerSave(
            POSIX_REASON_HARDWARE_FAIL);
    }

    PhysicalAction::ExecuteStepPulse
}
```

Listing 1: DOM-FAB-01 Physical Interlock Logic (Rust `#![no_std]`)

4 Laboratory Benchmark E: Physical Transduction HIL (`LAB-HIL-05`)

To extend the EDU-02 testing suite into physical hardware, EDU-06 defines Benchmark E: Physical Transduction HIL.

SPECIFICATION COMPLETE: ARC-SPEC-2026-EDU-06
DEPLOYMENT POSTURE: ARCHIVED FOR FUTURE PATH A EXTENSION
AUTHORIZATION HASH: A-77-DELTA-SHIELD-LOCKED

Table 2: LAB-HIL-05 Hardware Test Parameters

Test Metric	**Measurement Apparatus**	**Required Invariant Ceiling**
NMI Pin Latency	Logic Analyzer / Oscilloscope	From FAIL to pin state ≤ 11.99ms
Step Pulse Jitter (ΔJ)	Cycle Counter / Interlock	Timing jitter $\Delta J \leq 100$ns under load
FRAM Save Window	Capacitor Power Monitor	Save completion within $15 \ldots 25$ms zero-cross